EmbeddingsBench — findings

Text classification with `jinaai/jina-embeddings-v5-text-small-classification`: classifiers on frozen embeddings, TF-IDF, ensembles, LoRA fine-tuning and a German-native reference model on 18 English and German datasets.

Status: results as of 12 October 2026, 11:00 UTC. Fine-tuning of the remaining 10 datasets and the imbalance fine-tunes are still running (see the end); the report and the dashboard are updated when they finish. All accuracies are on held-out test splits; every setting (C, k, MLP shape, ensemble weight, fine-tuning checkpoint) was chosen on the validation split only.

Key findings

1. **Logistic regression is the right default on frozen embeddings.** Across 18 datasets a tuned MLP is on average +0.3 points better (median 0.0), an RBF SVM −0.9, kNN −6.5. A 40-trial MLP architecture search most often picks a linear model.
2. **The cheapest real gain is an ensemble with TF-IDF.** Averaging LogReg probabilities with a word-n-gram TF-IDF SVM beats LogReg on 13 of 18 datasets, +3.5 points on average, with no GPU training. It matches or beats LoRA fine-tuning on 4 of the 6 datasets where both were run.
3. **The best jina-based classifier is fine-tuned jina + TF-IDF**: +4.8 points over frozen LogReg on average (Yahoo! 72.9 %, arXiv 88.6 %, 20 Newsgroups 75.6 %, 10kGNAD 89.9 %). Fine-tuning and lexical features are complementary.
4. **After fine-tuning, the classifier on top no longer matters.** LogReg, RBF SVM and a tuned MLP on the fine-tuned encoder's embeddings land within about 1 point of its jointly trained linear head (the MLP search again chose a linear model on all six datasets). kNN trails LogReg by up to 20 points on frozen embeddings, but only by 0–6 points after fine-tuning, because fine-tuning pulls each class together.
5. **For German, a German-native encoder fine-tuned in full is stronger.** `gbert-large` beat LoRA-fine-tuned jina on all three German datasets: 10kGNAD 91.5 vs 88.6 % (89.9 % with TF-IDF), Amazon reviews 64.7 vs 63.6 %, MASSIVE 88.0 vs 87.0 %. It needed only about 20 min and 3.7 GB of GPU memory per dataset (8-bit AdamW).
6. **Frozen embeddings struggle with many fine-grained classes.** With 342 food categories (Open Food Facts DE) LogReg reaches 44.9 % while TF-IDF reaches 57.8 % and the ensemble with character n-grams 60.6 %. For 408 patent classes (HUPD): 50.2 / 52.3 / 54.3 %. Intent datasets with 60–150 classes are the exception (87–95 %), because their classes differ in meaning, not in vocabulary.
7. **Long, technical or legal texts favour lexical features.** On Swiss court decisions, arXiv and LEDGAR, TF-IDF alone beats LogReg on frozen embeddings by 3–7 points and is within 1.5 points of the best method.
8. **Class weighting buys rare-class recall; the price grows with the number of classes.** Tail recall (the third of classes with the fewest training examples): MTOP 27.9 → 36.8 % at equal accuracy, Swiss decisions 79.4 → 87.7 % (−1.5 points accuracy), Open Food Facts 36.4 → 46.8 % (−2.4), HUPD 12.6 → 29.8 % (−5.9).
9. **Zero-shot (label-name similarity) only works when label names are descriptive**: 91 % on emotions, 82 % on banking intents, under 10 % on legal areas, patents and product categories.
10. **2,048-token input only matters when most texts are long** (Swiss decisions +2.2, 10kGNAD +0.9, arXiv +0.5 points vs 512 tokens); for fine-tuning, head + tail truncation helped more than 1,024 tokens (10kGNAD 89.6 vs 88.8 %). **Matryoshka truncation to 512 dimensions costs 1.1 points on average** (at most 3.1), 256 dimensions 2.9 points.

Setup

- **Encoder:** `jinaai/jina-embeddings-v5-text-small-classification` (revision `4447914`), last-token pooling, L2-normalised 1,024-d embeddings, prompt `Document:`, up to 2,048 tokens.
- **Classifiers on frozen embeddings:** zero-shot (cosine similarity to label-name embeddings), kNN (cosine, k tuned), logistic regression (C tuned), RBF SVM, MLP (1–2 GELU layers) and a 40-trial MLP architecture search.
- **Text baselines:** TF-IDF (word 1–2-grams, sublinear tf) with a linear SVM; character 2–5-gram TF-IDF.
- **Ensembles:** weighted average of LogReg probabilities and softmaxed TF-IDF scores; weight and temperature chosen on validation.
- **Fine-tuning:** LoRA rank 16 on all attention and MLP projections, linear head initialised from the frozen LogReg, 3 epochs, 512 tokens (variants: 1,024 tokens, head + tail truncation, class-weighted and logit-adjusted loss), plus an ensemble with TF-IDF and the full classifier sweep on the fine-tuned embeddings. `deepset/gbert-large` fully fine-tuned as a German reference (8-bit AdamW, effective batch 16, 3 epochs).
- **Metrics:** accuracy, balanced accuracy, macro-F1, MCC, Cohen's κ, top-3 accuracy and tail recall (recall on the third of classes with the fewest training examples).
- **Hardware:** one RTX 3060 (12 GB), shared with other services.
- **Method categories** (as grouped in the dashboard): frozen jina + classifier · frozen jina with imbalance schemes · text only (TF-IDF) · ensembles of frozen embeddings and TF-IDF · fine-tuned jina (LoRA) · classifiers on fine-tuned jina embeddings · other model (gbert-large).

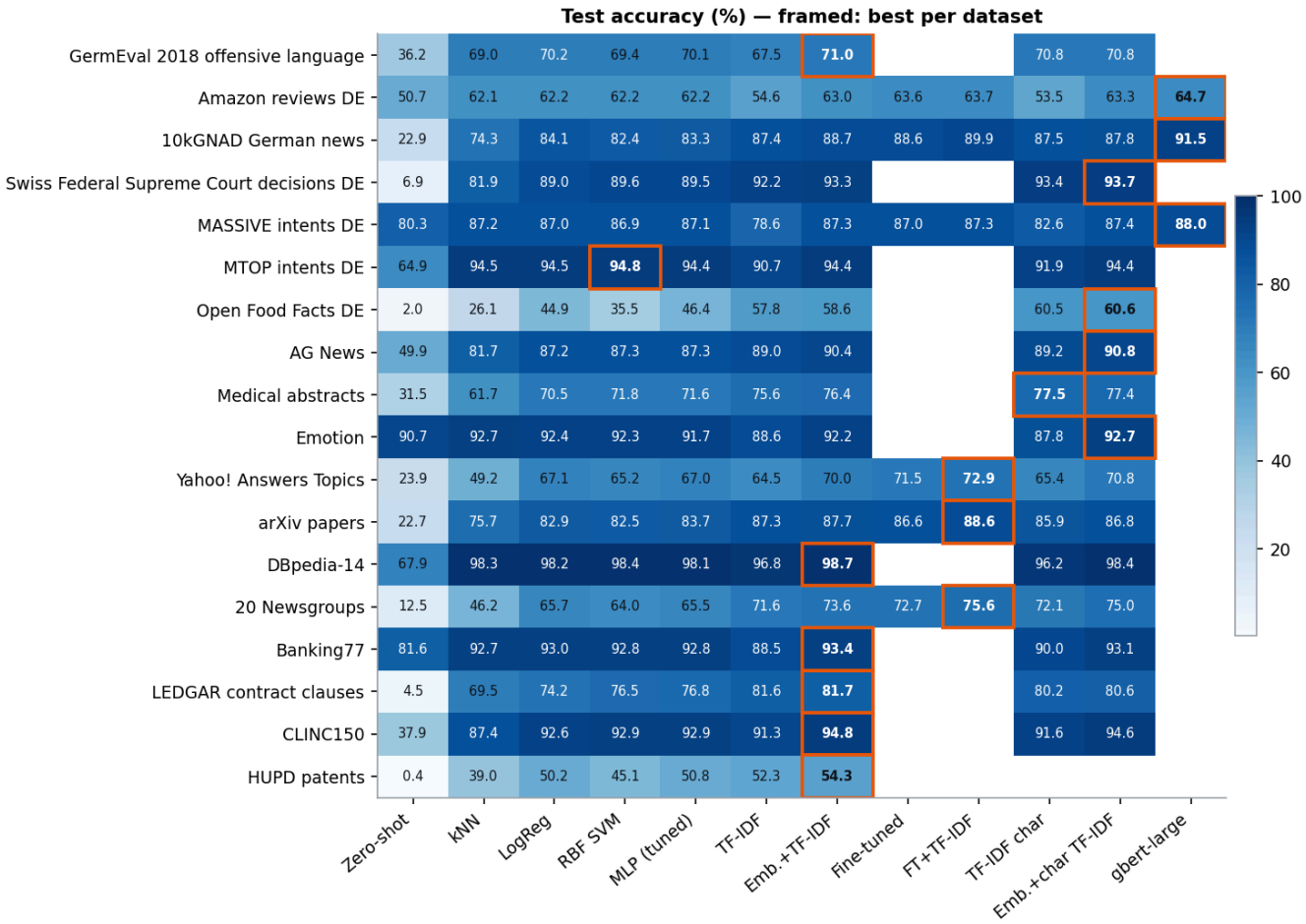
Datasets

Dataset	Lang	Domain	Classes	Train / val / test	Avg tokens	Train imbalance
GermEval 2018 offensive language	DE	tweets (offensive language)	4	4,258 / 751 / 3,398	45	47:1
Amazon reviews DE (stars)	DE	product reviews (sentiment)	5	10,000 / 5,000 / 4,999	65	1:1
10kGNAD German news	DE	news	9	7,860 / 1,384 / 1,027	732	3:1
Swiss Federal Supreme Court decisions DE (legal area)	DE	court decisions	47	25,737 / 3,002 / 3,002	3413	19:1
MASSIVE intents DE	DE	assistant intents	60	11,514 / 1,943 / 2,845	12	202:1
MTOP intents DE	DE	assistant intents (long tail)	103	11,072 / 1,956 / 1,756	14	1175:1
Open Food Facts DE (product categories)	DE	food products	342	69,366 / 8,790 / 8,790	18	12:1
AG News	EN	news	4	10,000 / 2,000 / 7,599	54	1:1
Medical abstracts	EN	medical	5	9,817 / 1,221 / 1,878	274	3:1
Emotion (tweets)	EN	emotion (tweets)	6	16,000 / 1,995 / 1,986	22	9:1
Yahoo! Answers Topics	EN	community Q&A	10	10,000 / 2,000 / 5,000	127	1:1
arXiv papers (11 categories)	EN	scientific papers	11	8,800 / 2,446 / 2,441	4300	1:1

Dataset	Lang	Domain	Classes	Train / val / test	Avg tokens	Train imbalance
DBpedia-14	EN	encyclopedia	14	7,000 / 1,400 / 2,800	79	1:1
20 Newsgroups	EN	forum posts	20	9,361 / 1,644 / 7,299	275	2:1
Banking77	EN	banking intents	77	8,497 / 1,496 / 3,069	15	5:1
LEDGAR contract clauses (LexGLUE)	EN	legal contracts	100	19,092 / 9,914 / 9,855	144	9:1
CLINC150 (in-scope)	EN	assistant intents	150	15,000 / 2,997 / 4,498	11	1:1
HUPD patents (IPC subclass)	EN	patents (IPC subclasses, long tail)	408	98,195 / 18,934 / 18,901	151	50:1

Splits are fixed and stratified where the source has none; large sources are capped per class (see `bench/data.py`). Train imbalance is the ratio of the largest to the smallest class in the training split.

Overview

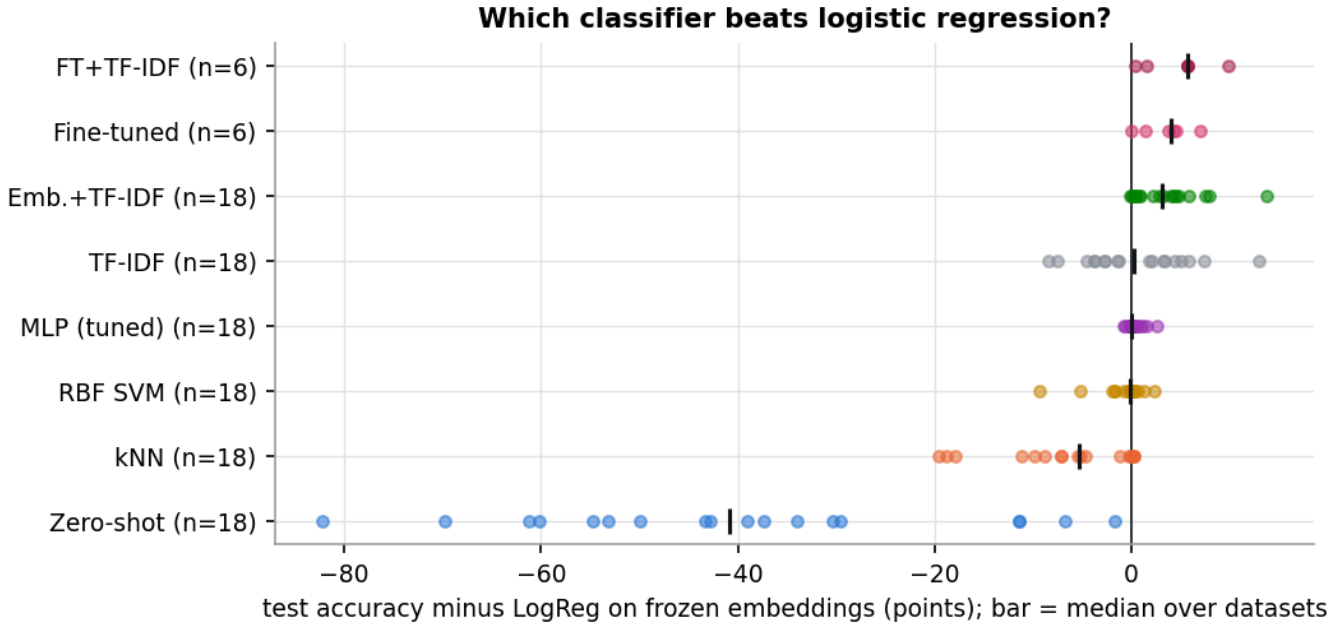


Dataset	Classes	Zero-shot	kNN	LogReg	MLP	TF-IDF	Emb.+TF-IDF	Best (method)	Macro-F1 of best
GermEval 2018 offensive language	4	36.2	69.0	70.2	70.3	67.5	71.0	**71.0** (Emb.+TF-IDF)	34.9
Amazon reviews DE	5	50.7	62.1	62.2	62.0	54.6	63.0	**64.7** (gbert-large)	64.6

Dataset	Classes	Zero-shot	kNN	LogReg	MLP	TF-IDF	Emb.+TF-IDF	Best (method)	Macro-F1 of best
10kGNAD German news	9	22.9	74.3	84.1	83.7	87.4	88.7	**91.5** (gbert-large)	91.5
Swiss Federal Supreme Court decisions DE	47	6.9	81.9	89.0	89.0	92.2	93.3	**93.7** (Emb.+char TF-IDF)	89.4
MASSIVE intents DE	60	80.3	87.2	87.0	87.0	78.6	87.3	**88.0** (gbert-large)	84.1
MTOP intents DE	103	64.9	94.5	94.5	94.6	90.7	94.4	**94.8** (RBF SVM)	83.2
Open Food Facts DE	342	2.0	26.1	44.9	45.4	57.8	58.6	**60.6** (Emb.+char TF-IDF)	56.9
AG News	4	49.9	81.7	87.2	86.9	89.0	90.4	**90.8** (Emb.+char TF-IDF)	90.7
Medical abstracts	5	31.5	61.7	70.5	72.0	75.6	76.4	**77.5** (TF-IDF char)	77.2
Emotion	6	90.7	92.7	92.4	92.3	88.6	92.2	**92.7** (Emb.+char TF-IDF)	88.1
Yahoo! Answers Topics	10	23.9	49.2	67.1	67.6	64.5	70.0	**72.9** (Fine-tuned jina + TF-IDF)	72.5
arXiv papers	11	22.7	75.7	82.9	82.8	87.3	87.7	**88.6** (Fine-tuned jina + TF-IDF)	87.9
DBpedia-14	14	67.9	98.3	98.2	98.2	96.8	98.7	**98.7** (Emb.+TF-IDF)	98.7
20 Newsgroups	20	12.5	46.2	65.7	65.5	71.6	73.6	**75.6** (Fine-tuned jina + TF-IDF)	74.0
Banking77	77	81.6	92.7	93.0	92.1	88.5	93.4	**93.4** (Emb.+TF-IDF)	93.3
LEDGAR contract clauses	100	4.5	69.5	74.2	75.1	81.6	81.7	**81.7** (Emb.+TF-IDF)	77.6
CLINC150	150	37.9	87.4	92.6	92.3	91.3	94.8	**94.8** (Emb.+TF-IDF)	94.8
HUPD patents	408	0.4	39.0	50.2	50.4	52.3	54.3	**54.3** (Emb.+TF-IDF)	38.7

The best method per dataset is picked by test accuracy for this overview; each method's own settings were chosen on validation. On GermEval 2018 and the long-tail sets (MTOP, HUPD), macro-F1 is far below accuracy: the majority classes dominate accuracy.

Classifiers on top of the embeddings

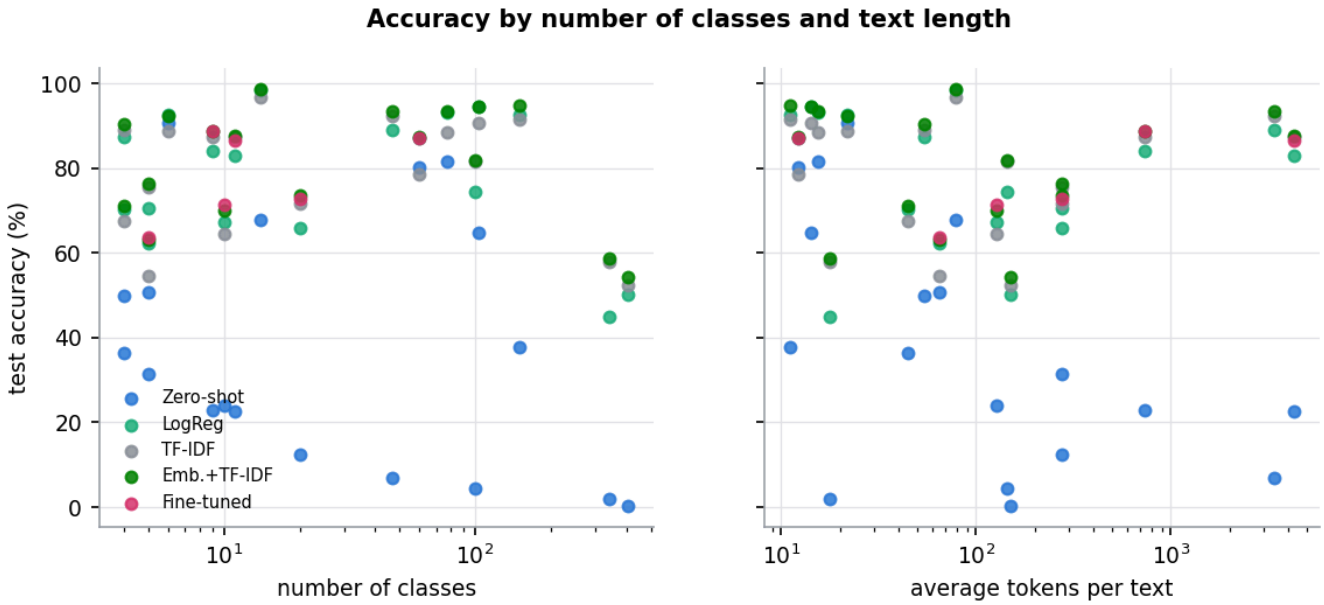


Test accuracy difference to LogReg on frozen embeddings, in points:

Classifier	Mean Δ (pts)	Median Δ (pts)	Better by > 0.5 pts	Datasets
FT+TF-IDF	+4.8	+5.7	5	6
Emb.+TF-IDF	+3.5	+3.0	13	18
Fine-tuned	+3.5	+4.0	5	6
TF-IDF	+0.5	+0.3	9	18
MLP (tuned)	+0.3	+0.0	5	18
RBF SVM	-0.9	-0.1	3	18
kNN	-6.5	-5.3	0	18
Zero-shot	-39.9	-40.9	0	18

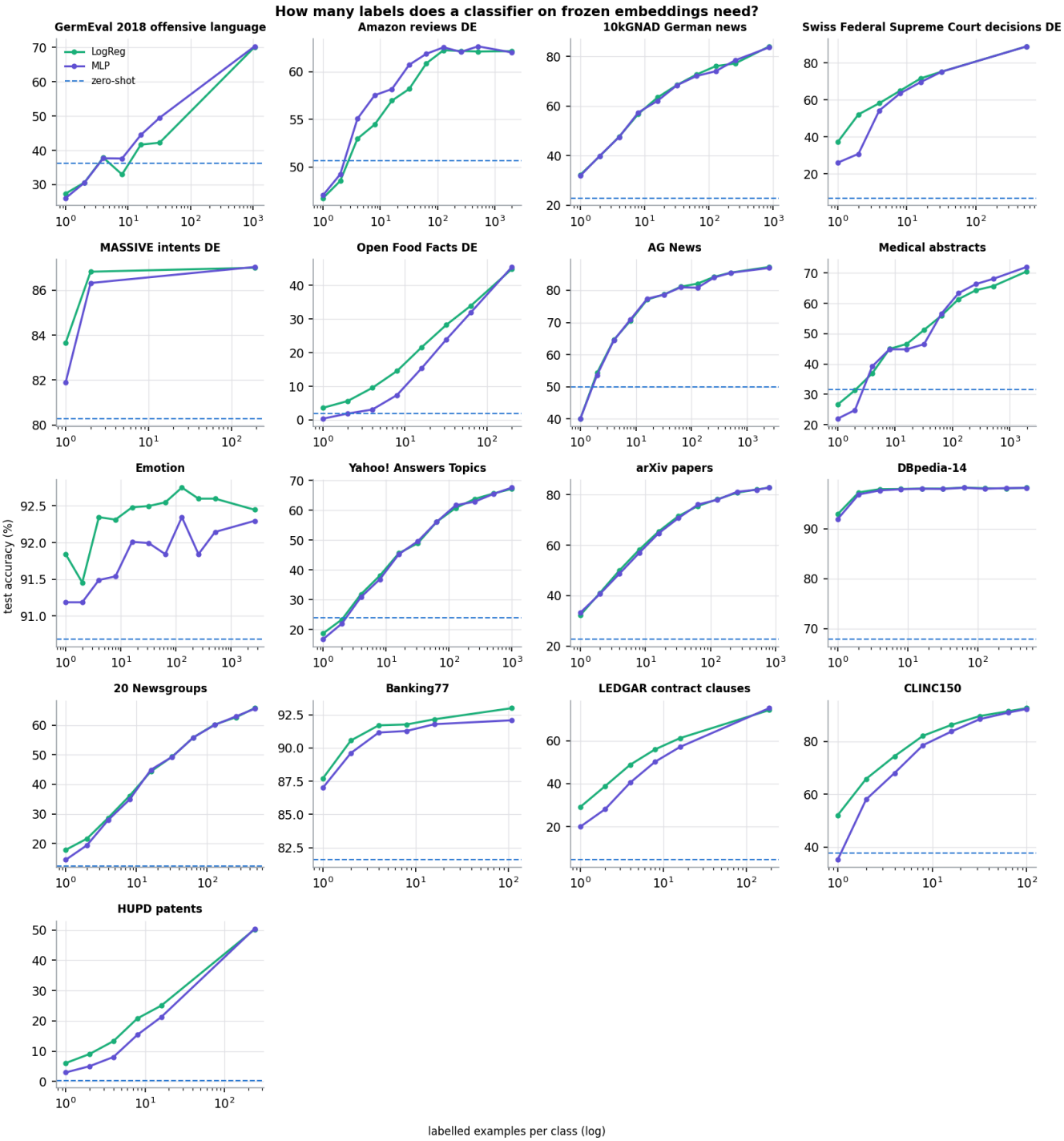
- **LogReg vs MLP:** the embeddings are already linearly separable where they separate at all. MLPs win by 1–2 points only on a few datasets with larger training sets (LEDGAR +2.6 tuned, Medical abstracts +1.1) and lose on small or many-class sets; the search mostly chose depth 0.
- **RBF SVM:** never clearly better than LogReg, slower to train (a 30k subsample was needed above 30k training texts) and on Open Food Facts 9 points worse.
- **kNN:** competitive on intents and emotions (within 0.5 points of LogReg on Banking77, MTOP, MASSIVE, Emotion), weak on topical and many-class data (−18 points on Yahoo!, −19 on Open Food Facts).
- **Zero-shot:** useful as a no-label starting point only where label names describe the class well.

Number of classes and text length



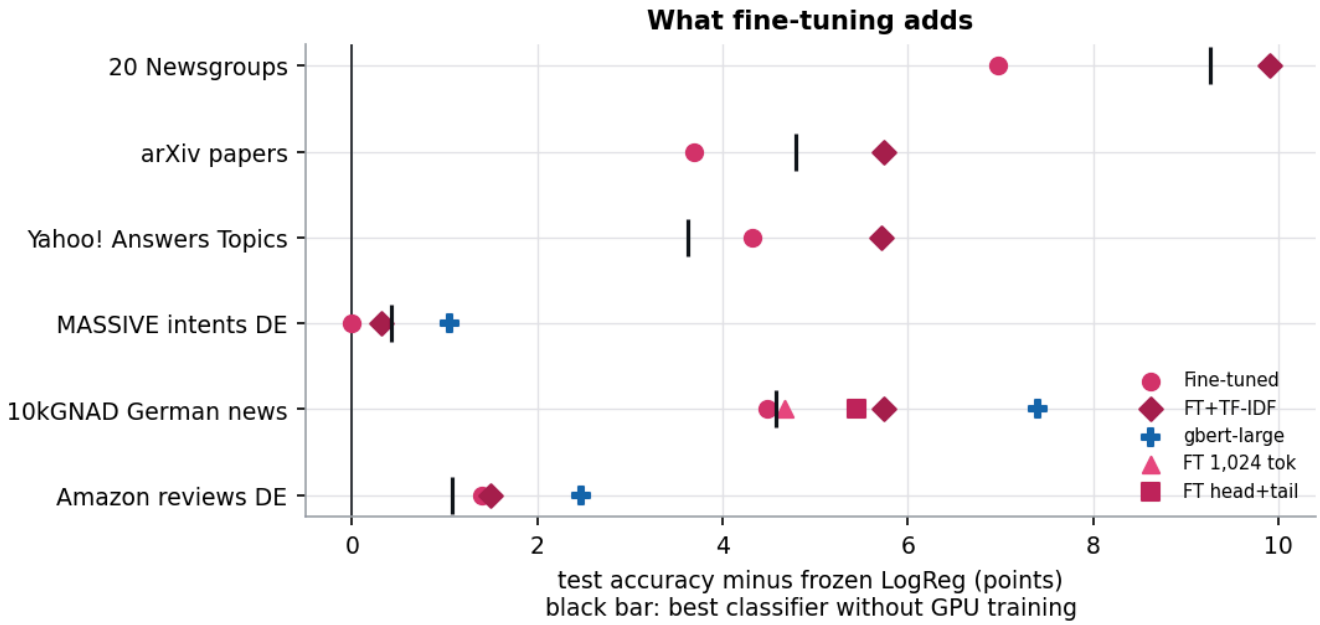
Accuracy on frozen embeddings drops sharply above about 150 classes, while TF-IDF degrades less: a single 1,024-d embedding appears to blur fine distinctions between neighbouring product or patent categories that specific words and word parts still separate. Text length alone is not the driver: short intent texts are classified well and short product names badly.

How many labels are needed



With 8–16 examples per class, LogReg on frozen embeddings already reaches most of its full-data accuracy on intent, emotion and encyclopedia data (Banking77 ~92 % at 16 per class, DBpedia ~98 % at 4). Topical and many-class datasets keep improving up to the full training set; there, more labels pay off more than a different classifier. LogReg is mostly ahead of the MLP at small sample sizes.

Fine-tuning

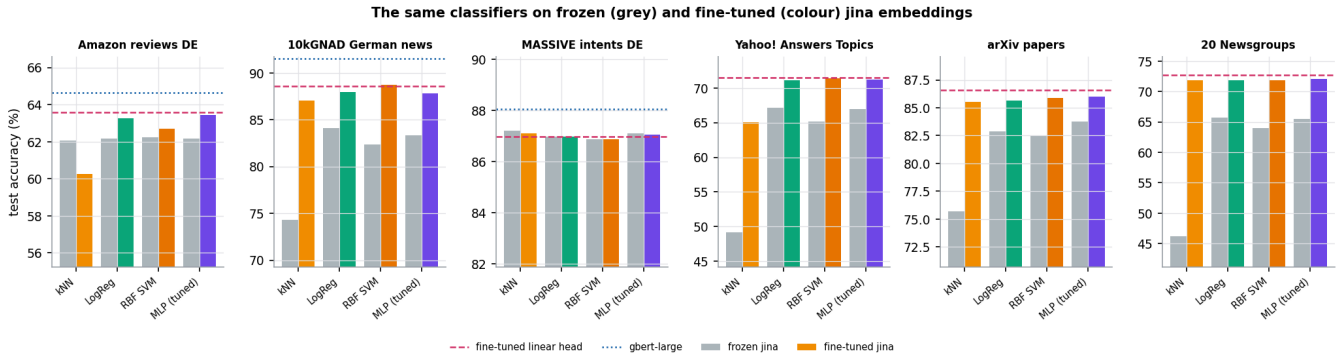


Dataset	Frozen LogReg	Fine-tuned	Best without GPU training	FT + TF-IDF	gbert-large	Train minutes
Amazon reviews DE	62.2	63.6	63.3	63.7	64.7	37
10kGNAD German news	84.1	88.6	88.7	89.9	91.5	76
MASSIVE intents DE	87.0	87.0	87.4	87.3	88.0	26
Yahoo! Answers Topics	67.1	71.5	70.8	72.9	–	39
arXiv papers	82.9	86.6	87.7	88.6	–	150
20 Newsgroups	65.7	72.7	75.0	75.6	–	46

- Fine-tuning pays off where classes are topical and the frozen embedding is not tuned to the distinction (Yahoo!, 20NG, news). On MASSIVE the best checkpoint was step 0, i.e. the untouched encoder.
- Fine-tuned jina alone beats the best classifier without GPU training on only 2 of 6 datasets; **combined with TF-IDF it is ahead on 5 of 6** (+0.4 to +2.1 points).
- 10kGNAD variants: head + tail truncation (first 384 + last 128 tokens) 89.6 %, 1,024 tokens 88.8 %, 512 tokens 88.6 %.
- `gbert-large` (335M parameters, all weights trained) is the best model on every German dataset where it ran. It is a different encoder, so its predictions cannot reuse stored jina embeddings.

Classifiers on fine-tuned embeddings

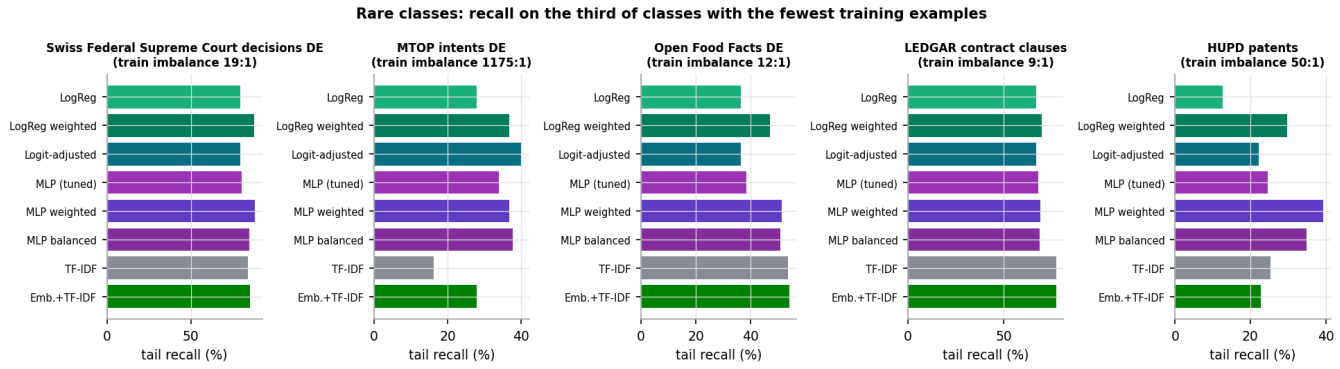
Does it pay to run the classifier sweep again after fine-tuning? The fine-tuned encoder's pooled, normalised embeddings were computed for train / val / test, and the same classifiers with the same grids were fitted on them.



Dataset	Frozen kNN	Frozen LogReg	Frozen RBF SVM	Frozen MLP (tuned)	FT-emb kNN	FT-emb LogReg	FT-emb RBF SVM	FT-emb MLP (tuned)	Fine-tuned	gbert-large
Amazon reviews DE	62.1	62.2	62.2	62.2	60.3	63.3	62.7	63.5	63.6	64.7
10kGNAD German news	74.3	84.1	82.4	83.3	87.0	87.9	88.7	87.8	88.6	91.5
MASSIVE intents DE	87.2	87.0	86.9	87.1	87.1	87.0	86.9	87.1	87.0	88.0
Yahoo! Answers Topics	49.2	67.1	65.2	67.0	65.1	71.1	71.4	71.2	71.5	–
arXiv papers	75.7	82.9	82.5	83.7	85.5	85.6	85.9	86.0	86.6	–
20 Newsgroups	46.2	65.7	64.0	65.5	71.8	71.8	71.9	72.1	72.7	–

- LogReg, RBF SVM and MLP on the fine-tuned embeddings end up within about 1 point of the fine-tuned linear head; none is consistently better. kNN is level on MASSIVE, 20NG and arXiv, but 3–6 points lower on Amazon and Yahoo!. The MLP search picked a linear model on all six datasets. **Keep the jointly trained linear head.**
- kNN profits most from fine-tuning (+13 to +26 points over frozen kNN on 10kGNAD, Yahoo! and 20NG). It becomes a reasonable choice when classes change often, since new labelled examples can be added without retraining. On sentiment (Amazon) it stays weakest.
- Caveat: the encoder saw the training texts, so the training embeddings are tighter than unseen ones; the test texts were never seen.

Class imbalance



Accuracy / balanced accuracy / macro-F1 / tail recall in % (tail recall: recall on the third of classes with the fewest training examples).

Swiss Federal Supreme Court decisions DE (legal area)

Classifier	Accuracy	Balanced acc.	Macro-F1	Tail recall
LogReg	89.0	83.9	81.4	79.4
LogReg weighted	87.5	86.1	79.6	87.7
Logit-adjusted	89.0	83.9	81.4	79.4
MLP (tuned)	89.5	83.6	80.5	80.4
MLP weighted	86.7	84.4	78.1	88.2
MLP balanced	87.8	85.1	80.3	84.9
TF-IDF	92.2	86.6	86.0	83.9
Emb.+TF-IDF	93.3	88.0	87.5	85.3

MTOP intents DE

Classifier	Accuracy	Balanced acc.	Macro-F1	Tail recall
LogReg	94.5	81.8	80.6	27.9
LogReg weighted	94.5	86.0	82.0	36.8
Logit-adjusted	93.7	87.0	78.4	40.2
MLP (tuned)	94.4	83.9	81.9	33.8
MLP weighted	93.8	85.5	79.0	36.8
MLP balanced	94.4	86.4	83.9	37.7
TF-IDF	90.7	69.8	67.1	16.2
Emb.+TF-IDF	94.4	81.7	80.5	27.9

Open Food Facts DE (product categories)

Classifier	Accuracy	Balanced acc.	Macro-F1	Tail recall
LogReg	44.9	42.4	42.8	36.4
LogReg weighted	42.5	46.2	42.3	46.8
Logit-adjusted	44.9	42.4	42.8	36.4
MLP (tuned)	46.4	43.4	43.7	38.3
MLP weighted	41.6	47.5	41.9	51.4
MLP balanced	42.3	47.7	43.0	50.7
TF-IDF	57.8	56.5	55.1	53.4
Emb.+TF-IDF	58.6	56.9	55.7	53.9

LEDGAR contract clauses (LexGLUE)

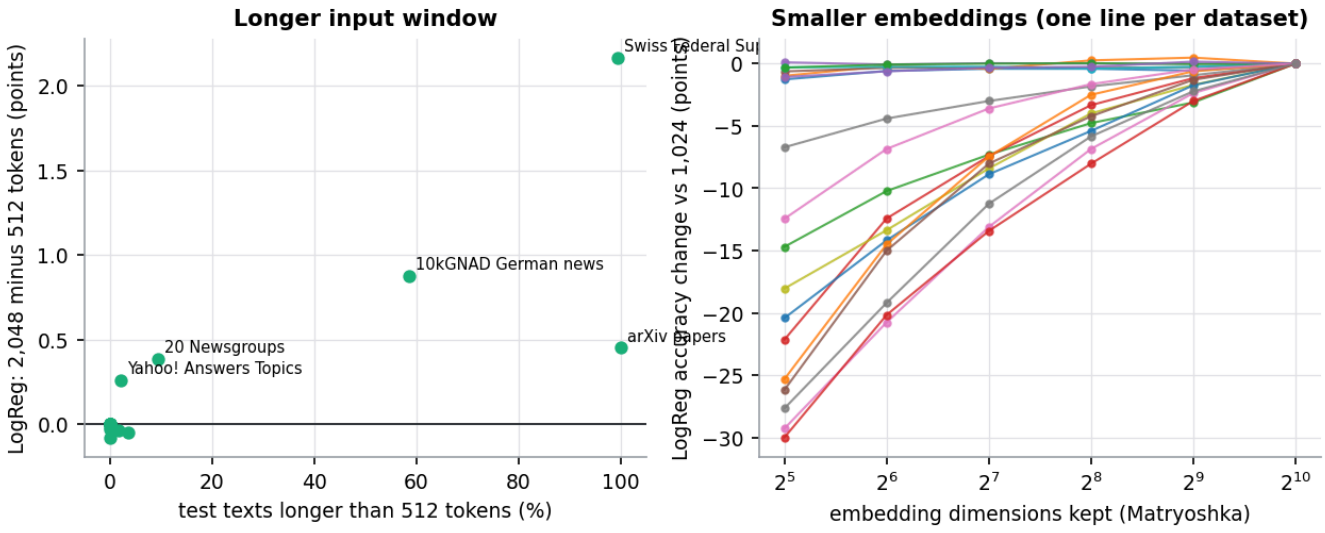
Classifier	Accuracy	Balanced acc.	Macro-F1	Tail recall
LogReg	74.2	71.5	68.3	67.0
LogReg weighted	74.1	72.7	68.8	69.8
Logit-adjusted	74.2	71.5	68.3	67.0
MLP (tuned)	76.8	72.6	69.7	68.1
MLP weighted	76.4	72.7	69.3	69.2
MLP balanced	74.5	71.8	68.4	68.8
TF-IDF	81.6	80.0	77.5	77.4
Emb.+TF-IDF	81.7	80.1	77.6	77.4

HUPD patents (IPC subclass)

Classifier	Accuracy	Balanced acc.	Macro-F1	Tail recall
LogReg	50.2	33.1	32.9	12.6
LogReg weighted	44.3	40.5	33.7	29.8
Logit-adjusted	48.8	38.4	34.3	22.2
MLP (tuned)	50.8	38.7	36.5	24.6
MLP weighted	40.7	42.8	31.5	39.3
MLP balanced	42.7	42.9	33.7	34.9
TF-IDF	52.3	39.6	38.3	25.3
Emb.+TF-IDF	54.3	39.2	38.7	22.7

- **Class-weighted LogReg** raises tail recall on every dataset: LEDGAR +2.8, MTOP +8.9, Swiss decisions +8.3, Open Food Facts +10.4, HUPD +17.2 points. The cost in accuracy grows with the number of classes and the imbalance (0 on MTOP and LEDGAR, −1.5 Swiss, −2.4 Open Food Facts, −5.9 HUPD).
- **Logit adjustment** (subtracting $\tau \cdot \log$ prior at prediction time) is free to apply after training. τ is chosen on validation: $\tau = 0$ (no change) on LEDGAR, Swiss and Open Food Facts; on MTOP and HUPD it improves balanced accuracy (+5.2, +5.3 points); on HUPD at a much smaller accuracy cost than weighting (−1.4 vs −5.9 points).
- **MLP with weighted loss or balanced batches** gives the highest tail recall of the embedding classifiers on Open Food Facts and HUPD, again at the cost of accuracy.
- TF-IDF is strongest on rare classes in the legal and product datasets, but on MTOP (1175:1, short texts) it is weakest: 16 % tail recall.
- Class-weighted and logit-adjusted *fine-tuning* on MTOP and Swiss decisions is running.

Input length and embedding size



The longer window helps only in proportion to how many texts exceed 512 tokens. Matryoshka truncation: 512 dimensions −1.1 points on average (worst −3.1), 256 −2.9, 128 −5.5. The loss below 256 dimensions varies widely between datasets (see the figure).

Cost

Embedding runs at about 11,000 tokens/s and LoRA fine-tuning at about 1,700 tokens/s on the shared RTX 3060; both scale with tokens, not with the number of classes. LogReg, kNN and TF-IDF train in seconds to minutes on CPU, except TF-IDF with character n-grams on large many-class sets (HUPD: more than 11 GB of RAM, skipped).

Recommendations

Default pipeline. Embed once, train LogReg (tune C on a validation split), and add a TF-IDF SVM in a probability ensemble. That is the best or within a point of the best on most datasets, costs one embedding pass, and retrains in minutes when labels change.

Larger datasets (≫ 10k training texts). - Embed once and keep the embeddings: classifier experiments then cost CPU minutes, not GPU hours. - Fit LogReg with a fast solver (SAGA or a GPU softmax layer) and a coarse C grid; subsample for SVMs and for hyper-parameter search (we used 30k for the RBF SVM). - An MLP becomes worth trying above roughly 20k training texts (LEDGAR +2.6 points), still with a 1-layer default. - Fine-tune only on a capped subset first (e.g. 1,000 per class): the learning curves show diminishing returns, and fine-tuning time grows linearly with tokens (about 1,700 tokens/s on an RTX 3060 for LoRA on jina v5 small). - Watch memory for TF-IDF on many classes: character n-grams on 98k texts × 408 classes needed more than 11 GB of RAM; use hashing with fewer features or word n-grams only. - Use 512-d Matryoshka embeddings when storage or kNN search matters; budget about 1 point of accuracy.

Many classes (> 100–150). - Do not rely on frozen embeddings alone: always include TF-IDF (word and character n-grams) and the ensemble; on 342 and 408 classes it was 4–16 points ahead of LogReg. - Report balanced accuracy, macro-F1 and top-3 alongside accuracy; macro-F1 on HUPD is 16 points below accuracy. - Make sure each class has enough examples (≥ 20, ideally ≥ 100); merge or drop catch-all and near-duplicate categories (as done for Open Food Facts). - Consider a hierarchy: predict the parent category first (e.g. the IPC class), then the subclass among its children. - Use class weighting, and check tail recall. - Zero-shot and kNN are not viable at this scale.

Short texts with semantic classes (intents, emotions). Frozen embeddings + LogReg are enough (87–95 %); kNN is a good alternative when classes change often, because no retraining is needed. Fine-tuning brings little.

Long documents. Use the 2,048-token window (or head + tail when fine-tuning) and combine with TF-IDF; legal and scientific texts carry much of their signal in vocabulary.

German. jina v5 handles German well on frozen embeddings and is the practical choice when one embedding store serves several tasks. When one task justifies fine-tuning, train `gbert-large`: it was 1–3 points ahead of LoRA-fine-tuned jina on all three German datasets, at lower training cost.

When to fine-tune at all. Fine-tune when classes are topical and the frozen + TF-IDF ensemble plateaus; then ensemble the fine-tuned model with TF-IDF again. Skip it for short intent and emotion texts, where the frozen encoder is already as good. After fine-tuning, keep the linear head; re-running the classifier sweep on the fine-tuned embeddings brought nothing.

Imbalanced data. Decide first whether accuracy or rare-class recall matters. Class-weighted LogReg costs nothing to try; logit adjustment can be tuned after training on validation; with hundreds of classes expect to trade several points of accuracy for large tail-recall gains.

Still running

- LoRA fine-tuning, fine-tuned + TF-IDF ensembles and the fine-tuned-embedding sweep on the 10 remaining datasets (GermEval 2018, Banking77, CLINC150, Emotion, DBpedia, AG News, Open Food Facts, Medical abstracts, LEDGAR; HUPD with 1 epoch).
- gbert-large on GermEval 2018, MTOP, Open Food Facts and Swiss decisions.
- Plain, class-weighted and logit-adjusted fine-tuning on MTOP and Swiss decisions.

Reproduce

Code, data preparation and all raw results are in this repository: `results/results.json` (every number above, incl. per-setting validation scores), `bench/` (pipeline), `bench/report.py` (these figures and tables), `docs/` (the interactive dashboard, also on GitHub Pages). See the README for setup and the GPU server configuration.